Data Communication & Computer Networks (CN)

Topic 5 of 7 : Transmission Medium

Curation in progress — 36 auto-extracted questions pending verification (see Pending Review at the end)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • RSMSSB IA • RSMSSB SA • Data Entry/DD • BCI • SCI • REET • RAS & other Rajasthan Govt. Exams**

Total Questions: **0** verified MCQs + **36** pending review

Question Type: MCQ — every question has 4 options (A, B, C, D)

Languages: **Bilingual:** English + Hindi (हिन्दी)

Every question includes: ✔ Correct answer (why it is RIGHT) • ★ Nukta / key point (exam tip) • ✖ why each wrong option is WRONG

Verification: Every answer cross-verified **4 times** (Forouzan facts + official answer keys of Subject Test / Unit Test Solutions where available)

Syllabus Coverage — sub-topic wise order

Sub-topic curation in progress — auto-extracted questions are listed in the **"Pending Review"** section at the end.

How to read this PDF: Each question shows 4 options (A-D) in English and Hindi. The line starting with ✔ gives the correct answer and why it is right. The line starting with ★ is the **Nukta** — the key point to remember for the exam. The lines starting with ✖ explain why every wrong option is wrong. Read the question, try it yourself, then check the answer.

System: This PDF is generated by **tools/cn_build_topic.py** from **tools/data/topic5_data.py**. To add more questions, drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions are auto-classified into topics and appended here (as pending) for review. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

These questions were auto-extracted & classified from uploaded PDFs. Answers/nukte to be verified before counting.

⚠ NEW — auto-ingested, answer needs verification (pending review)

P1. "Manchester Encoding" and "Differential Manchester Encoding" are signal encoding techniques used at which layer? "Manchester Encoding" "Differential Manchester Encoding"

(A) Physical Layer
(B) Data Link Layer
(C) Network Layer
(D) Presentation Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P2. Assertion

(A) : Fiber Optic cable is immune to Electromagnetic Interference (EMI). (A): (EMI) Reason (R): It transmits data in the form of light pulses rather than electrical signals. (R): (A) Both (A) and (R) are true, and (R) is the correct explanation of (A)
(B) Both (A) and (R) are true, but (R) is NOT the correct explanation.
(C) (A) is true, but (R) is false. /.
(D) (A) is false, but (R) is true.

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P3. A company needs long-distance communication (5–10 km), high data rate, and immunity from electromagnetic interference (EMI). Which transmission media is most suitable? (5–10), EMI transmission media

(A) UTP
(B) Coaxial cable
(C) Single-mode fiber
(D) Multi-mode fiber

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P4. In a copper cable network, attenuation increases when frequency increases. What change will reduce attenuation? frequency attenuation attenuation

(A) Increase cable length
(B) Use lower category UTP
(C) Use thicker copper core
(D) Increase signal frequency

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P5. A network uses coaxial cable but suffers from signal reflection. Which problem is most likely the cause? coaxial cable signal reflection

(A) EMI
(B) Incorrect termination resistance
(C) Crosstalk
(D) Low bandwidth

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P6. Which of the following statements are TRUE for fiber optic cables? I. Immune to tapping II. High bandwidth III. Requires electrical grounding IV. Works using total internal reflection

(A) I, II, IV
(B) II, III
(C) I, III, IV
(D) I, II

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P7. A UTP cable used in Gigabit Ethernet shows high delay skew. Which is the cause? Gigabit Ethernet UTP delay skew

(A) Different twist rates in cable pairs
(B) Excessive EMI
(C) Fiber splice error
(D) Wrong wireless channel

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P8. A channel has high bandwidth but suffers from high noise. Which encoding scheme will provide better error resistance? bandwidth noise - encoding error resistance

(A) NRZ
(B) Manchester
(C) AMI
(D) ASK

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P9. Which phase of a computer virus life cycle is responsible for remaining hidden until a specific condition is met? D H

(A) Propagation Phase
(B) Triggering Phase
(C) Dormant Phase / C (E)
(D) Execution Phase / F

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P10. Reduction in signal strength over distance is called: v %– B

(A) Distortion –
(B) Noise –
(C) Attenuation – L
(D) Interference – T

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P11. Which noise is caused due to electromagnetic coupling between nearby cables? ™S v - š›

(A) Thermal noise – Z
(B) Crosstalk –]
(C) Impulse noise – ^_
(D) Quantization noise – c

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P12. Which parameter of an analog signal represents the number of cycles per second? v ‰ ‡ 9

(A) Amplitude –
(B) Phase –
(C) Frequency – k
(D) Wavelength – pZ

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P13. Which impairment cannot be completely eliminated but only minimized? Ÿ B y

(A) Attenuation – L
(B) Distortion –
(C) Noise –
(D) Delay –

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P14. Which of the following is a wireless networking technology? / H I + K

(A) Ethernet
(B) Wi-Fi
(C) Fiber Optic / H
(D) DSL

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P15. What is the maximum data rate of a transmission medium known as? / ? Q

(A) Bandwidth / P
(B) Throughput / R
(C) Latency / T
(D) Jitter

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P16. Which cable has maximum bandwidth? ?**

(A) Coaxial
(B) Twisted pair
(C) Optical fiber
(D) USB cable

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P17. Which wireless standard provides the highest data rate

(A) 802.11b
(B) 802.11g
(C) 802.11n
(D) 802.11ac What is the purpose of DHCP Discover message?DHCP Discover (a) Request IP lease (b) Identify gateway (c) Detect MAC address (d) Announce server

[Source: CN Topic Test 6.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P18. What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences

(A) Unshielded twisted pair
(B) Optical fiber
(C) Coaxial cable
(D) Microwave

[Source: CN Topic Test-01.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P19. Which of the following cannot be used as a medium for 802.3 ethernet? 802.3

(A) A thin coaxial cable
(B) A twisted pair cable
(C) A microwave link
(D) A fiber optical cable

[Source: CN Topic Test-01.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P20. What technique is used to allow multiple signals to share the same communication channel in a network

(A) Multiplexing
(B) Phase modulation
(C) Analog modulation
(D) Digital modulation

[Source: CN Topic Test-02.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P21. What is the meaning of bandwidth in a network

(A) Transmission capacity of a communication channel
(B) Connected components in a network
(C) Class of IP used in Network
(D) Interconnected by communication channels

[Source: CN Topic Test-02.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P22. The period of a signal is 100ms. What is its frequency in kilohertz? 3. 100

(A) 10-1 kHz
(B) 10 -2 kHz
(C) 10-3 kHz
(D) 10-4 kHz

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P23. Which of the following is NOT a type of transmission media

(A) Coaxial cable
(B) Fiber optics
(C) NIC
(D) Twisted pair

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P24. In digital communication, what does "attenuation" refer to? " "

- **(A)** Signal delay
- **(B)** Signal loss
- **(C)** Signal interference
- **(D)** Signal amplification

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P25. Which of the following best explains why fiber optic cables have a lower bit error rate than copper cables

- **(A)** Higher signal power
- **(B)** Immunity to electromagnetic interference
- **(C)** Simpler modulation techniques
- **(D)** Larger physical size

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P26. Given a bit rate of b bits/sec, the time required to send 16 bits is: b / , 16 required :

- **(A)** 16*b sec
- **(B)** 16/b sec
- **(C)** 16b sec
- **(D)** b16 sec

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P27. Which of the following transmission media works on the principle of total internal reflection

- **(A)** Optical fiber cable
- **(B)** Shielded twisted pair cable
- **(C)** Unshielded twisted pair cable
- **(D)** coaxial cable

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P28. An analog signal carries 4 bits in each signal unit. if 1000 signal units are sent per second, then baud rate and bit rate of the signal are 4 1000

- **(A)** 1000 bauds/sec, 4000 bps
- **(B)** 1000 bauds/sec, 500 bps
- **(C)** 4000 bauds/sec, 1000 bps
- **(D)** 2000 bauds/sec, 1000 bps

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P29. Crosstalk is typically minimized in twisted pair cables by:

(A) Increasing the power level
(B) Increasing the twist rate
(C) Adding shielding to the core
(D) Using longer cables

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P30. What is the typical maximum length of an unshielded twisted pair (UTP) cable in Ethernet before signal degradation occurs? ( )

(A) 50 meters
(B) 100 meters
(C) 150 meters
(D) 200 meters

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P31. Which connector is commonly used for coaxial cable in physical networks

(A) RJ-45
(B) USB-C
(C) BNC
(D) HDMI

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P32. Which type of wireless transmission requires line-of-sight

(A) Wi-Fi
(B) Infrared
(C) Radio
(D) Bluetooth

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P33. The effective bandwidth of a communication channel is primarily limited by:

(A) Protocol overhead
(B) Noise and attenuation
(C) File size
(D) MAC address filtering

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P34. Which of the following statements is not true with respect to microwaves

(A) Electromagnetic waves with frequencies from 300 GHz to 400 Thz.

(B) Propagation is line-of-sight.

(C) Very high-frequency waves cannot penetrate walls.

(D) Use of certain portions of the band requires permission from authorities. 28 Optical fiber uses reflection to guide light through a channel, in which angle of incidence is____the critical angle. _____ (a) A equal to (b) less than (c) greater than (d) less than or equal to

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P35. Which of the following transmission media is used in Bluetooth Technology: :

(A) Radio links

(B) Microwave links

(C) CVSAT Communication

(D) Fiber-optic

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P36. The primary use of scramblers in Line Coding Schemes is to -

(A) make the data look random

(B) compress transmission data

(C) increase signal bandwidth

(D) decrease signal bandwidth

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 5 / Signals & Performance]